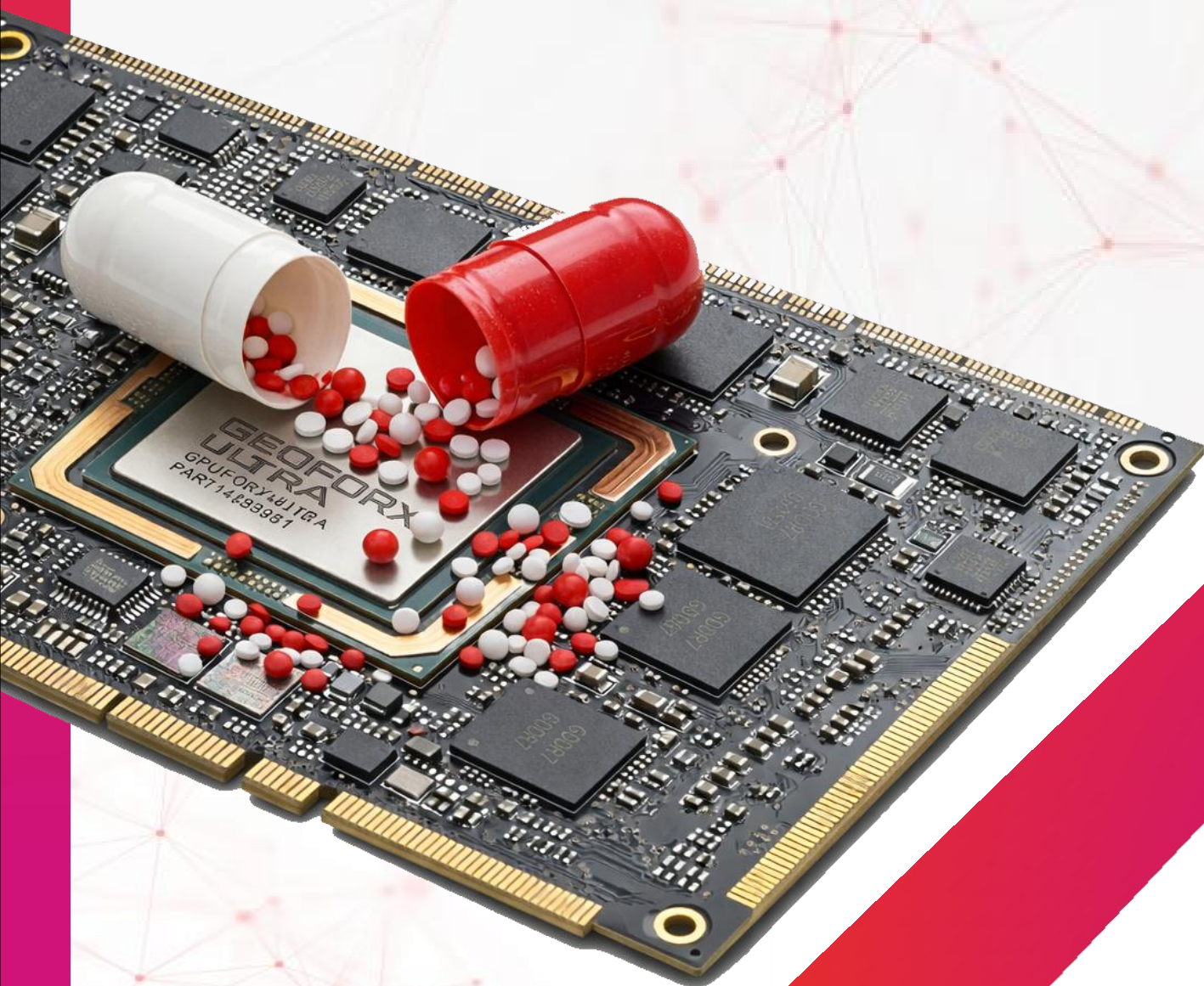


Large Language Models Are Rewriting Pharma's Playbook



LLMs are rewriting pharma's playbook — here's exactly how

Large Language Models have moved from experimental curiosity to strategic infrastructure across the pharmaceutical industry, with 173+ AI-discovered drug programs now in clinical development and every top-20 pharma company actively deploying LLM-based solutions.

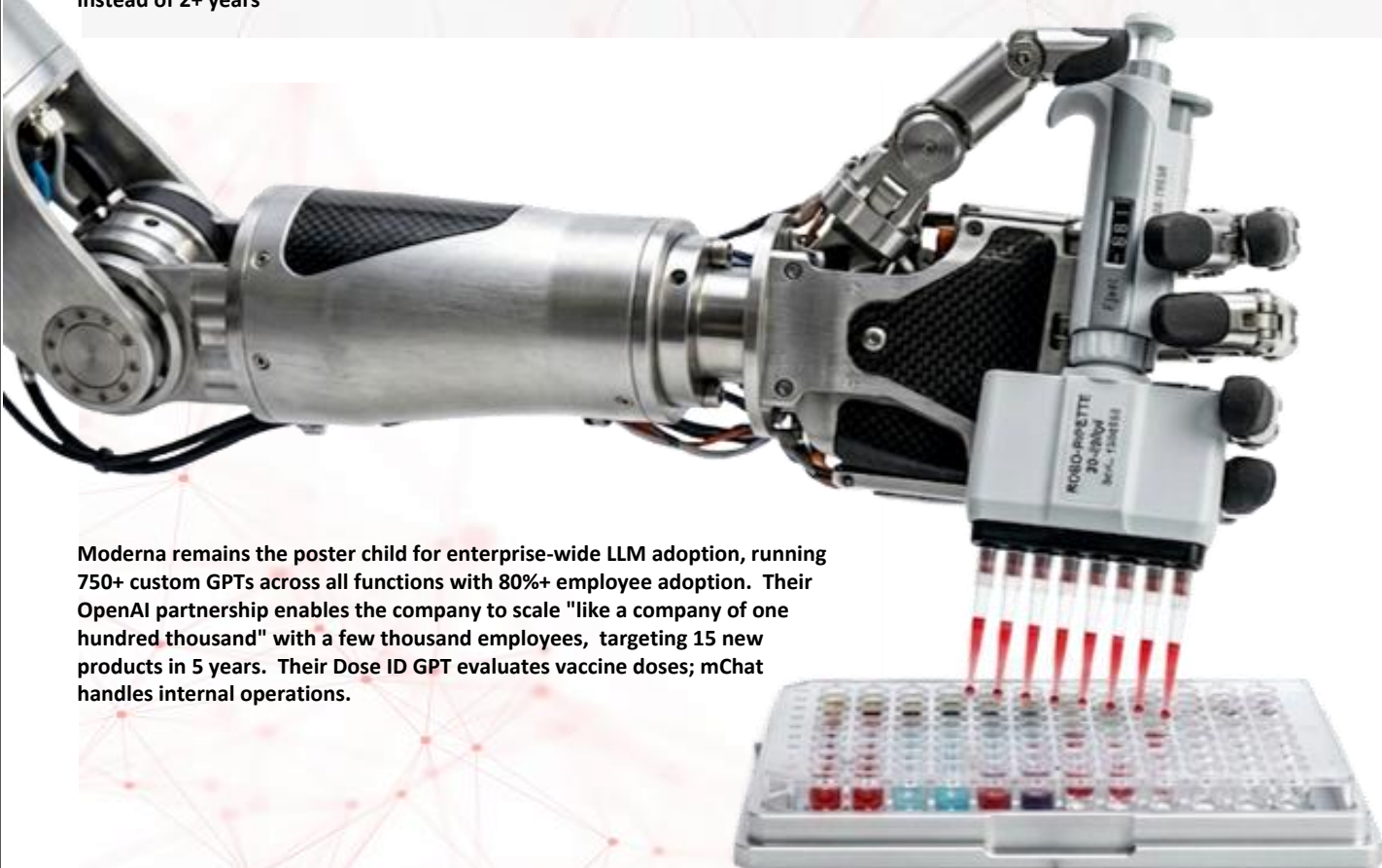
The AI-in-pharma market reached \$1.94–4.35 billion in 2025 and is growing at 27–42% CAGR, with McKinsey estimating \$60–110 billion in annual economic value potential. Yet a critical reality check remains: no AI-discovered drug has received FDA approval as of early 2026. Insilico Medicine's rentosertib — the first drug where both the target and the molecule were discovered using generative AI — published its Phase IIa results in Nature Medicine (June 2025), representing the field's most advanced clinical proof-of-concept. The industry sits at an inflection point where billions in investment meet the biological constraints that no algorithm can shortcut.

Which pharma giants are betting on LLMs — and how much

Nearly every major pharmaceutical company has made significant LLM commitments in 2025–2026, but the scale and ambition vary dramatically. Eli Lilly leads in pure investment, announcing a \$1 billion NVIDIA co-innovation AI lab (January 2026) alongside a \$2.75 billion deal with Insilico Medicine (March 2026) for AI-designed oral therapeutics. Lilly also partnered with Isomorphic Labs (up to \$1.7 billion in milestones) and built what it calls "the most powerful supercomputer owned by a pharmaceutical company."

Roche/Genentech deployed 3,500+ NVIDIA Blackwell GPUs on-premises — the largest GPU footprint in pharma — with an estimated hardware investment exceeding \$65 million. Their "Lab-in-the-Loop" strategy integrates AI into nearly 90% of eligible small-molecule programs, delivering one oncology degrader molecule 25% faster and a backup molecule in 7 months instead of 2+ years

Moderna remains the poster child for enterprise-wide LLM adoption, running 750+ custom GPTs across all functions with 80%+ employee adoption. Their OpenAI partnership enables the company to scale "like a company of one hundred thousand" with a few thousand employees, targeting 15 new products in 5 years. Their Dose ID GPT evaluates vaccine doses; mChat handles internal operations.



Sanofi aspires to become "the first pharma company powered by AI at scale," partnering with OpenAI and Formation Bio to build custom drug development software, launching the Muse clinical trial recruitment tool, and achieving \$300 million in supply chain savings from AI-driven efficiencies. AstraZeneca acquired AI startup Modella AI (January 2026), committed \$200 million with Tempus AI and Pathos AI for an oncology foundation model, and reports >50% faster target drug design.

Other notable moves include Novartis expanding its Isomorphic Labs partnership (up to \$1.2 billion in milestones, February 2025) and its \$1 billion Generate:Biomedicines deal for AI-designed protein therapeutics; Merck deploying GPTeal across 50,000 employees and reducing clinical study report drafting from 2–3 weeks to 3–4 days using McKinsey's

QuantumBlack AI; GSK licensing Noetik's cancer foundation models for \$50 million upfront;

BioNTech operating the Kyber near-exascale supercomputer with 240+ AI/ML professionals from its InstaDeep acquisition (£362 million);

and Amgen developing the open-source AMPLIFY protein language model that tripled protein engineering speed and halved discovery timelines.

A cross-industry consortium trend has also emerged: Bristol-Myers Squibb, Takeda, AbbVie, J&J, and Astex launched the Federated OpenFold3 Initiative (October 2025) to share proprietary protein-small molecule structural data for AI model training.

The LLM technology stack powering drug discovery

The pharma LLM ecosystem spans general-purpose models, domain-specific biological models, and protein language models, each serving distinct functions across the drug development pipeline.

General-purpose LLMs (GPT-4/GPT-5, Claude, Gemini) dominate enterprise operations. AstraZeneca's "AZ ChatGPT" runs on Azure OpenAI with 12,000 employees completing genAI certifications by April 2025. Pfizer's "Charlie" platform provides secure GPT access for marketing, R&D, and manufacturing. GPT-5, launched August 2025, is described as "a legitimate Ph.D.-level expert" in healthcare applications. Anthropic launched Claude for Life Sciences (October 2025) with connectors to Benchling, PubMed, and Medidata, followed by the \$400 million acquisition of Coefficient Bio (April 2026) for molecular biology capabilities.

Protein language models represent pharma's most transformative LLM category. ESM3 (EvolutionaryScale, published in Science, January 2025) is a 98-billion-parameter generative model trained on 2.78 billion proteins that generated a novel green fluorescent protein equivalent to simulating 500 million years of evolution. AlphaFold 3 (DeepMind, May 2024) predicts molecular structures 50% more accurately than traditional methods, while Isomorphic Labs' proprietary successor IsoDDE doubles AF3's accuracy — described as "an AlphaFold 4" in a June 2025 Nature article. Recursion's Boltz-2 (June 2025) achieves state-of-the-art binding affinity prediction at speeds 1,000× faster than free energy perturbation calculations.

Domain-specific pharma models fill specialized niches. Google's TxGemma (March 2025) offers three model sizes (2B, 9B, 27B parameters) trained on 7 million examples, outperforming specialized models on 64 of 66 key drug discovery tasks including blood-brain barrier crossing and binding affinity prediction. MedGemma (Google I/O 2025) provides open-source healthcare AI supporting 3D CT/MRI and histopathology. NVIDIA BioNeMo expanded significantly (January 2026) with new Clara open models for RNA structure prediction (RNAPro) and drug synthesizability assessment (ReaSyn v2), plus Merck's KERMT model for ADMET prediction pretrained on ~11 million molecules. Insilico Medicine's Chemistry42 generates drug candidates from scratch — rentosertib moved from AI-designed molecule to clinical trials in under 30 months versus the typical 5+ years.

The MAGE antibody language model (Vanderbilt, published in Cell, November 2025) generated functional antibodies against novel H5N1 influenza strains, while PharmBERT (trained on 138,924 drug labels) outperforms existing biomedical models in adverse drug reaction detection.



AI Adoption Across Top Pharma — A Snapshot

In 2025–2026, every top pharmaceutical company has placed significant bets on Large Language Models — but the scale is staggering. Eli Lilly alone committed over \$5 billion across partnerships with NVIDIA, Insilico Medicine, and Isomorphic Labs. Roche deployed 3,500+ GPUs — the largest in pharma. Moderna runs 750+ custom GPTs with 80%+ employee adoption. Sanofi saved \$300 million through AI-driven supply chain optimization. AstraZeneca, Novartis, Merck, BioNTech, and Amgen each built proprietary AI platforms spanning drug design to clinical reporting. The message is clear: LLMs are no longer experimental — they are core infrastructure reshaping how medicines are discovered, developed, and delivered. With 173+ AI-discovered drugs now in clinical trials and multi-billion-dollar deals becoming routine, the pharmaceutical industry's AI transformation has passed the point of no return.



Recursion

AI Partner: NVIDIA / Exscientia
Country: USA
Technology: Boltz-2 Phenom ML
Live Product: 65PB Dataset 3T+ relations
Investment: \$150M+
Therapeutics: Multi therapeutic



**Insilico
Medicine**

AI Partner: Internal Platform
Country: HKG/ USA
Technology: Chemistry42 PandaOmics
Live Product: Rentosertib (Phase IIa)
Investment: \$293M IPO
Therapeutics: IPF Oncology



AI Partner: Azure OpenAI
Country: USA
Technology: Charlie Platform
Live Product: Charlie Internal
Investment: Undisc
Therapeutics: Vaccines Oncology

AstraZeneca

AI Partner: Absci / Modella / Tempus AI
Country: UK
Technology: A2 ChatGPT Azure OpenAI
Live Product: Internal 12K trained
Investment: \$200M+
Therapeutics: Oncology Biomarkers

Lilly

AI Partner: NVIDIA / Insilico / Isomorphic
Country: USA
Technology: BioNeMo Chemistry42
Live Product: AI Drug Lab
Investment: \$5B+
Therapeutics: Small Mol Oral Drugs

MERCK

AI Partner: McKinsey QB OpenAI
Country: USA
Technology: GPTeal AI CSR Writer
Live Product: GPTeal (50K users)
Investment: Undisc.
Therapeutics: Clinical Reports



AI Partner: Noetik AI
Country: UK
Technology: Cancer Foundation Model
Live Product: Oncology Pipeline
Investment: \$50M+ (upfront)
Therapeutics: Oncology Immunology

moderna

AI Partner: OpenAI GPT-4
Country: USA
Technology: 750+ Custom GPTs
Live Product: Dose ID GPT mChat
Investment: Undisc.
Therapeutics: mRNA Vaccines

Roche
Genentech

AI Partner: NVIDIA BioNeMo
Country: SWI
Technology: 3500+ GPUs Blackwell
Live Product: Lab-in-Loop
Investment: \$65M+ (HW)
Therapeutics: Oncology Degradors

BIONTECH

AI Partner: InstaDeep (acquired)
Country: DEU
Technology: Kyber Super computer
Live Product: AI Neoantigen Pipeline
Investment: \$500 M+
Therapeutics: mRNA Cancer Tx

sanofi

AI Partner: OpenAI / Formation Bio
Country: France
Technology: Muse Platform Plaid
Live Product: Muse Trial Recruitment
Investment: \$300M savings
Therapeutics: Immunology Oncology

NOVARTIS

AI Partner: Isomorphic Labs / Generate Bio
Country: SWI
Technology: AlphaFold Protein LMs
Live Product: AI Protein Design
Investment: \$2.2B+ (deals)
Therapeutics: Protein Tx Small Mol

AMGEN

AI Partner: NVIDIA / Internal AI
Country: USA
Technology: AMPLIFY Protein LM
Live Product: Open-source AMPLIFY
Investment: Undisc.
Therapeutics: Antibodies Bispecifics

LLM Integration across Pharma Workflow Stage

Target identification has become AI's proving ground

LLMs have transformed target discovery from literature-heavy manual curation to automated multi-omics reasoning. Insilico Medicine's PandaOmics platform — now serving 13 of the top 20 global pharma companies — combines NLP-based literature mining, knowledge graphs, and omics data integration with ChatPandaGPT for conversational target queries. PandaOmics identified TNIK as a novel anti-fibrotic target for IPF, which became the foundation of rentosertib (ISM001-055), the first drug with both an AI-discovered target and AI-designed molecule.

Microsoft's BioGPT, trained on 15M+ PubMed abstracts, enables hypothesis generation for drug-target interactions and drug repurposing. Geneformer, an LLM trained on 30 million single-cell transcriptomes, models disease states for therapeutic target identification.

BenevolentAI's Knowledge Graph maps billions of relationships between diseases, proteins, genes, and clinical outcomes — most famously identifying baricitinib as a COVID-19 treatment within 48 hours in January 2020, which later received FDA Emergency Use Authorization after showing a 38% reduction in mortality.

AlphaFold 2/3 (Isomorphic Labs/DeepMind) has predicted structures of 200M+ proteins, enabling structure-based target identification. Recursion's Boltz-2, an open-sourced billion-parameter generative model for protein 3D structure prediction, was downloaded by 40,000+ users within weeks of its 2025 release. The DrugLM framework integrates LLM embeddings into drug-target interaction models, achieving a 38.5% improvement over baseline.

Key 2025–2026 deals in target identification include Isomorphic Labs' combined ~\$3B partnerships with Eli Lilly (\$45M upfront, up to \$1.7B milestones) and Novartis (\$37.5M upfront, up to \$1.2B milestones), with Novartis doubling its commitment from 3 to 6 projects in July 2025. Recursion's partnership with Roche/Genentech carries >\$12B in potential milestones across up to 40 programs.

Generative chemistry and virtual screening are redefining hit discovery

Chemistry42 (Insilico Medicine) uses reinforcement learning with 460+ medicinal chemistry filters for de novo molecular design. It demonstrated synergy with AlphaFold2 for CDK20 inhibitor design, producing compounds with Kd values of 210 nM and 375 nM. Schrödinger's FEP+ platform can screen billions of compounds in one week; for an EGFR project, it explored 23 billion designs and identified 4 novel scaffolds in 6 days. For WEE1, it achieved >10,000x selectivity over PLK1 and reached a clinical candidate in 10 months versus an industry average of 4–6 years.

Novartis used generative AI to computationally design 15 million potential compounds for a Huntington's disease molecular glue degrader, then worked with only ~60 molecules in the lab to arrive at a potent brain-penetrant scaffold. A 2025 Nature Communications study using a VAE with nested active learning cycles generated drug-like molecules, synthesized 9, and found 8 with in vitro activity including one with nanomolar potency.

Rentosertib's development exemplifies the end-to-end promise: target identified via PandaOmics, molecule designed via Chemistry42, preclinical candidate achieved in 18 months at approximately 10% of conventional cost. Phase IIa results showed a dose-dependent 98.4 mL mean FVC improvement versus a 62.3 mL decline in placebo for IPF patients.

AI is reshaping preclinical safety and ADMET prediction

Unexpected toxicity causes ~30% of drug development failures, and ~90% of clinical candidates never reach market, with 25% of failures stemming from safety issues not predicted by animal testing. The FDA in 2025 launched an initiative to phase out traditional animal testing, encouraging AI-powered models and New Approach Methodologies.

Inductive Bio received up to \$21M from ARPA-H (December 2025) for AI drug toxicity models using organoids and ex-vivo human tissue. The POSTECH GPD Model (published in eBioMedicine, 2025) improved AUPRC from 0.35 to 0.63 and AUROC from 0.50 to 0.75 for predicting clinical toxicity versus chemical-data-only approaches. Novartis's Data42 unit generated cardiac activity profiles predicting clinical cardiac toxicity, identifying molecules with strong cardiosafety risk signals validated in the lab. Over 20 ADMET prediction platforms were catalogued in a 2025 Briefings in Bioinformatics review.

Clinical trial AI is producing measurable efficiency gains

TrialGPT (NIH/NLM) reduces trial pools by >90%, matches patient eligibility at 87.3% accuracy, and enables clinicians to spend 40% less time screening. RECTIFIER, tested in the COPILOT-HF trial, achieved 97.9–100% accuracy across 13 eligibility criteria with 92.3% sensitivity and 93.9% specificity. Tempus AI's TIME Program spans ~100 sites with 5M+ cancer patients and 1,500+ active trials; its Patient Query AI agent screened out ~72% of ineligible patients across 8 trials.



Unlearn.AI's digital twins — using Conditional Restricted Boltzmann Machines trained on 1M+ longitudinal clinical records — received EMA qualification of their PROCOVA method for Phase 2/3 trials. In an AbbVie Alzheimer's study, digital twins could reduce total sample sizes by 9–15% and control arms by 17–26% while maintaining statistical power. Sanofi's Muse platform, built with OpenAI and Formation Bio, represents the first purpose-built AI application for pharma clinical operations.

Regulatory submissions are being compressed from weeks to days

Merck's AI-powered Clinical Study Report authoring platform, developed with McKinsey QuantumBlack, reduced first-draft time from 2–3 weeks to 3–4 days and cut CSR drafting hours from 180 to 80 (56% reduction) while reducing errors by 50%. Merck's GPTeal enterprise LLM platform serves 50,000+ employees monthly. The AutoIND platform reduces initial nonclinical summary drafting time by ~97% with zero critical regulatory errors.

Filing one month faster for a \$1B asset unlocks ~\$60M in NPV according to McKinsey. Leading companies are achieving filings 8–12 weeks after database lock, a 50–65% timeline reduction versus 2020. The FDA published draft guidance in January 2025 establishing a 7-step credibility framework for AI tools in drug development, and issued joint FDA-EMA guiding principles in January 2026.

Manufacturing, pharmacovigilance, and commercial AI are scaling rapidly

In manufacturing, Pfizer's AI reduced cycle time by 67% and unplanned maintenance downtime by 20%. Digital twin-based predictive maintenance yields 15–25% improvements in equipment reliability. SAP Joule Agents cut procurement cycle times by 22% in pharma operations.

In pharmacovigilance, Oracle Argus processes 10M+ cases/year with ~60% market share. ArisGlobal's Advanced Signals module delivers up to 80% faster signal assessment. Tech Mahindra launched an agentic AI PV solution with NVIDIA NeMo in 2025. Genpact's AI-driven case processing cuts PV costs by up to 40% and reduces report drafting from days to hours.

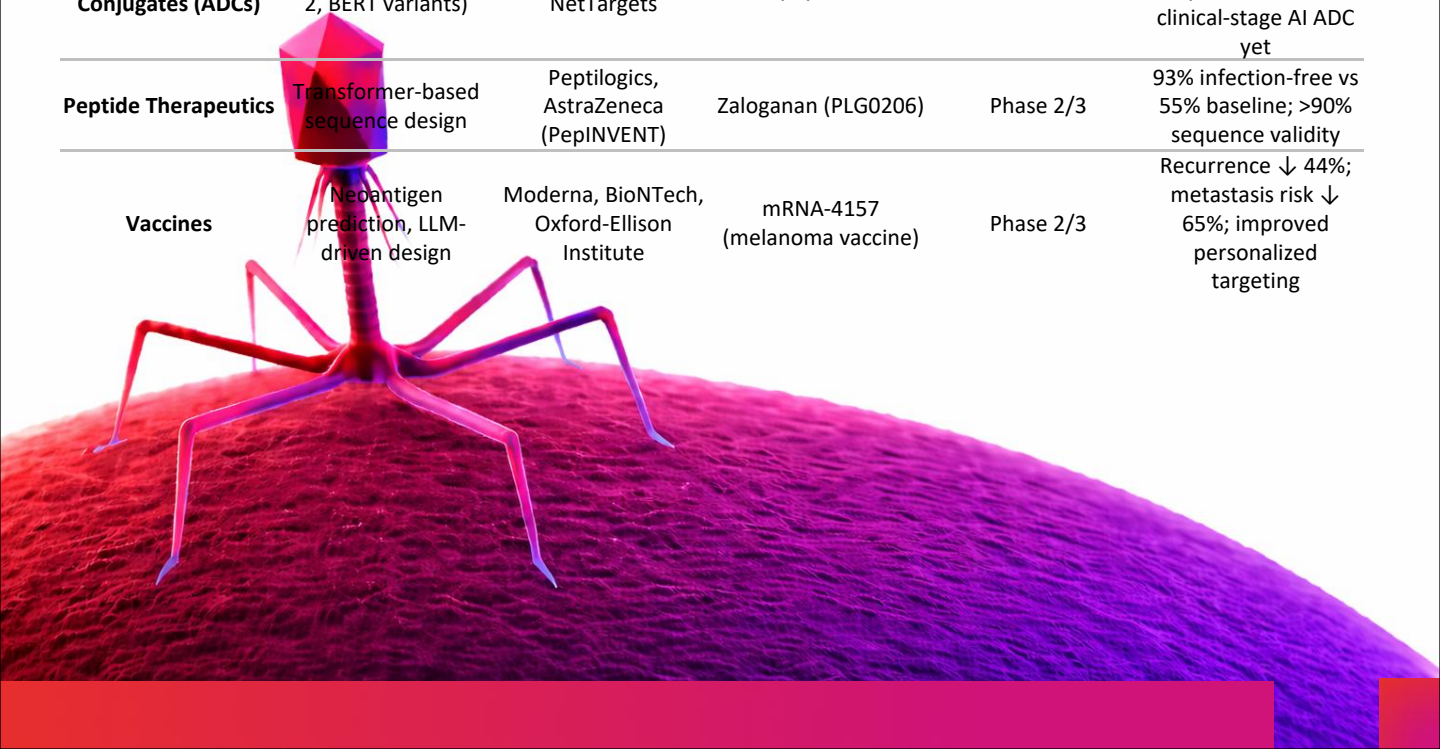
In commercial operations, Veeva's AI for PromoMats (December 2025) uses Anthropic and Amazon models for MLR review — Moderna reports this "moves us closer to nearly touch-free MLR." Veeva's CRM AI Agents serve Bristol Myers Squibb, Roche, Novo Nordisk, Gilead, and Merck. Pfizer's "Charlie" GPT instance, deployed to hundreds of marketers, aims for 3–5x content creation volume. McKinsey estimates generative AI could deliver \$18–\$30B annually in pharma commercial value.

Stage	Key Technologies / Models	Companies / Platforms	Key Achievements	Quantitative Impact
Target Identification	NLP, Knowledge Graphs, Multi-omics AI	Insilico (PandaOmics), Microsoft BioGPT, BenevolentAI	AI-discovered targets like TNIK; COVID drug repurposing (baricitinib)	Faster discovery; billions of relationships mapped
Protein Structure Prediction	AlphaFold 2/3, Generative protein models	DeepMind, Recursion	200M+ protein structures predicted	Massive structural database acceleration
Hit Discovery & Design	Generative chemistry, RL, VAEs	Chemistry42, Schrödinger FEP+	Billions of compounds screened; novel scaffolds identified	10 months vs 4–6 years (traditional)
Preclinical (ADMET & Toxicity)	AI toxicity models, organoids	Inductive Bio, Novartis Data42	Better toxicity prediction vs animal models	AUPRC ↑ 0.35 → 0.63; AUROC ↑ 0.50 → 0.75
Clinical Trials	LLM matching, digital twins	TrialGPT, Unlearn.AI, Tempus	Patient matching, reduced trial size	Screening time ↓ 40%, sample size ↓ up to 15%
Regulatory Submission	AI document generation	Merck (GPTeal), AutoIND	CSR automation	Time ↓ 2–3 weeks → 3–4 days
Manufacturing	Digital twins, predictive AI	Pfizer, SAP	Process optimization	Cycle time ↓ 67%, downtime ↓ 20%
Pharmacovigilance	AI signal detection	Oracle Argus, ArisGlobal	Faster safety signal detection	Cost ↓ up to 40%
Commercial Operations	AI marketing & MLR automation	Veeva, Pfizer GPT	Automated content generation	3–5x content output

LLM Application Across Therapeutic Modalities

This table highlights how LLMs and generative AI are transforming different therapeutic modalities with varying maturity levels and impact. Small molecules remain the most advanced and commercially active area, with over 173 clinical programs and significant reductions in discovery timelines. Biologics, particularly antibodies, show dramatic efficiency gains with up to 100-fold improvements in hit rates due to protein language models. Emerging areas like mRNA and gene therapy demonstrate substantial optimization in expression, stability, and vector engineering through transformer-based architectures. Meanwhile, modalities such as ADCs, cell therapy, and peptides are still evolving but already show promising improvements in design accuracy and screening efficiency. Overall, AI is proving most effective in design, optimization, and early-stage development, while later-stage clinical validation continues to be the key bottleneck across all modalities.

Modality	Key AI / LLM Technologies	Leading Platforms / Companies	Representative Examples	Development Stage	Quantitative Impact / Outcomes
Small Molecules	Generative chemistry, physics-based modeling, RL	Insilico (Chemistry42), Schrödinger (FEP+), Recursion OS	Zasocitinib (TYK2 inhibitor)	Phase 3	173+ AI drugs in pipeline; timeline reduced from 3–4 years → 13–18 months; 30–40% faster discovery
Antibodies (Biologics)	Protein language models, generative design	Chai Discovery (Chai-2), Generate Biomedicines, Absci	GB-0895 (anti-TSLP), ABS-101	Phase 3 / Clinical	16–20% hit rate (vs ~0.1% earlier); ~100× improvement; 86% developable candidates
mRNA Therapeutics	Transformer models, sequence optimization	GEMORNA, mRNA-GPT, Moderna AI stack	mRNA-4157 (cancer vaccine)	Phase 2/3	Expression ↑ up to 150×; stability ↑ 5×; antibody titers ↑ 128×
Gene Therapy	CRISPR LLMs, protein engineering	CRISPR-GPT, OpenCRISPR-1, Dyno Therapeutics	AI-designed Cas proteins, AAV capsids	Preclinical / Early clinical	Directed evolution reduced from 3–6 cycles (3+ years) → 1–2 cycles (~1 year); expression ↑ 15×
Cell Therapy	AI-driven CAR-T design, screening models	St. Jude, Somite AI	CAR construct optimization	Preclinical	~1000 CAR constructs screened in days; improved tumor clearance (4/5 mice)
Antibody-Drug Conjugates (ADCs)	Multi-modal AI (ESM-2, BERT variants)	ADCNet, Debiopharm, NetTargets	Dual-payload ADCs	Preclinical	Improved multi-component optimization; no clinical-stage AI ADC yet
Peptide Therapeutics	Transformer-based sequence design	Peptilogics, AstraZeneca (PepINVENT)	Zaloganan (PLG0206)	Phase 2/3	93% infection-free vs 55% baseline; >90% sequence validity
Vaccines	Neoantigen prediction, LLM-driven design	Moderna, BioNTech, Oxford-Ellison Institute	mRNA-4157 (melanoma vaccine)	Phase 2/3	Recurrence ↓ 44%; metastasis risk ↓ 65%; improved personalized targeting



Who leads the AI arms race serving pharma

This section highlights leading AI-driven pharma companies building end-to-end discovery platforms, integrating multi-omics data, generative models, and physics-based simulations. It emphasizes how large-scale data, proprietary AI pipelines, and billion-dollar partnerships are accelerating early-stage drug discovery and candidate generation.

Company	HQ./ Founded	AI Approach Type	Core Platforms / Models	Data Scale / Infrastructure	Pipeline & Key Assets	Clinical Stage	Key Deals / Partnerships	Financials	Strengths	Limitations / Risks	Strategic Position
 RELAY THERAPEUTICS	USA / 2016	Protein dynamics AI	Dynamo®	Cryo-EM + MD simulations	Zovagalisib	Phase 3	—	\$555M cash	Dynamic protein modeling	Narrow focus	Precision oncology leader
Generate Biomedicines	USA / 2018	Generative protein AI	Generate Platform™	Large protein datasets	GB-0895	Phase 3	Angen (\$1.9B)	\$805M funding	De novo protein design	Clinical validation ongoing	Biologics innovator
 Atomwise	USA / 2012	Deep learning screening	AtomNet®	3T compound library	TYK2 inhibitor	Preclinical	Sanofi (> \$1B)	\$219M funding	Massive search space	Limited clinical success	Screening pioneer
 Healex	UK / 2014	AI drug repurposing	Healnet	Literature + clinical data	HLX-1502	Phase 2	Sanofi	~\$120M	Rare disease focus	Smaller scale	Niche specialist
 OWKIN	France/US / 2016	Federated learning AI	MELLODDY, K Pro	Distributed hospital data	Biomarker discovery	Clinical support	Sanofi (\$270M)	\$354M funding	Privacy-preserving AI	Complex coordination	Collaborative AI leader
 UNLEARN	USA / 2017	Digital twin AI	PROCOVA	Longitudinal clinical data	Trial simulation	Clinical	AbbVie, J&J	\$135M funding	Reduces trial size	Regulatory adoption risk	Clinical trial disruptor
 Iambic	USA / 2019	Multimodal transformer	Enchant™, NeuralPlexer	Protein-ligand datasets	IAM1363	Phase 1	Jazz Pharma	\$306M funding	Fast iteration cycles	Early-stage	Agile discovery platform
 Chai Discovery	USA / 2023	Generative protein AI	Chai-1, Chai-2	Structural biology datasets	Antibody discovery	Preclinical	—	\$1.3B valuation	High hit rates (~20%)	Very new company	High-potential disruptor
 Standigm	Korea / 2015	Integrated AI platforms	ASK™, BEST™, Insight™	Multi-source biomedical data	20+ pipelines	Preclinical	SK Chemicals	~\$70M	Fast lead generation	Limited global scale	Regional AI leader
VALO HEALTH	USA / 2019	Multi-omics AI	Opal platform	70T molecule database	Obesity, T2D pipeline	Preclinical/Clinical	Novo Nordisk (\$4.6B)	~\$500M funding	Massive data integration	Complex modeling	Big pharma partner

Who leads the AI arms race serving pharma

Cont.

Company	HQ./ Founded	AI Approach Type	Core Platforms / Models	Data Scale / Infrastructure	Pipeline & Key Assets	Clinical Stage	Key Deals / Partnerships	Financials	Strengths	Limitations / Risks	Strategic Position
 Insilico Medicine	HK & NY / 2014	End-to-end generative + LLM	PandaOmics, Chemistry42, Biology42, InClinico	Multi-omics + NLP + knowledge graphs	Rentosertib + 22 preclinical + 7 clinical	Phase IIa	Lilly (\$2.75B), Sanofi (\$1.2B), Servier (\$888M)	\$56M revenue, IPO \$293M	Full-stack AI pipeline	Clinical success still unproven	Vertical AI pharma leader
 Recursion	USA / 2013	Phenomics + ML + automation	Recursion OS 2.0	60+ PB biological imaging data	Multiple Phase 1/2 assets	Early clinical	Roche (>\$12B), Bayer, Merck	\$785M cash	Largest proprietary dataset	High failure (REC-994)	Data-centric discovery giant
 Somorphic Labs	UK / 2021	Structure-based AI (AlphaFold)	AlphaFold 3 + proprietary models	Protein structure datasets	Preclinical programs	Preclinical	Lilly + Novartis (~\$3B)	\$600M funding	Solves "undruggable" proteins	No clinical validation yet	Frontier research leader
 EvolutionaryScale	USA / 2023	Foundation model (biology)	ESM3 (98B params)	2.78B proteins, 10 ⁴ FLOPS	esmGFP protein	Preclinical	CZ Biohub acquisition	\$142M seed	Multimodal protein reasoning	Early-stage commercialization	Biology foundation model pioneer
 absci	USA / 2011	Generative biologics + wet lab AI	Integrated Drug Creation™	Billions of cell experiments/wk	ABS-201, ABS-101	Phase 1/2	AstraZeneca, Almirall	\$144M cash	Fast (6-week discovery)	Scaling validation risk	Rapid biologics generator
 Benevolent ^{AI}	UK / 2013	Knowledge graph AI	Disease-gene-drug graphs	Billions of relationships	BEN-8744 + 20+ programs	Clinical	AstraZeneca, Merck	Acquired (2025)	Explainable AI	Clinical failures (eczema trial)	Repurposing specialist
 Excscientia	UK / 2012	AI-driven chemistry	Precision design systems	Integrated chemistry datasets	DSP-1181	Clinical	Acquired by Recursion	—	First AI drug in trials	Integration dependency	Legacy pioneer
 Schrodinger	USA / 1990	Physics + AI hybrid	FEPl, Glide, LiveDesign	Molecular simulation at scale	SGR-1505, TAK-279	Phase 1–3	Novartis (\$2.3B)	\$257M revenue	High predictive accuracy	Expensive compute	Industry-standard platform
 TEMPUS	USA / 2015	Clinical + multimodal AI	TIME platform	Genomics + imaging + EHR	Clinical trial optimization	Clinical	AstraZeneca (\$200M)	\$1.27B revenue	Largest clinical dataset	Data privacy complexity	Clinical AI leader
 XtalPi	China / 2015	AI + quantum + robotics	Integrated discovery platform	Automated labs + quantum models	PEP08, SIGX1094R	Phase 1	DoveTree (\$5.99B)	404% growth	Hybrid physics-AI edge	High capital cost	Next-gen discovery stack
 NOETIK	USA / 2022	Foundation cancer model	OCTO-VC	200M+ cells	Biomarker prediction	Preclinical	GSK (\$50M)	\$62M funding	Virtual cell modeling	Early stage	Foundation model startup

Who leads the AI arms race serving pharma

Five technology providers have established dominant positions across different pharma segments, each with distinct competitive advantages.

NVIDIA controls the infrastructure layer, with no real competitor in pharma GPU compute. The BioNeMo platform serves 200+ organizations and spans the full stack from hardware to domain-specific microservices. Key pharma partnerships — Eli Lilly (\$1B), Roche (3,500+ GPUs), Thermo Fisher, IQVIA, Illumina — make NVIDIA the indispensable backbone of pharma AI.

Google DeepMind / Isomorphic Labs dominates structural biology. AlphaFold's lineage gives them unmatched protein structure prediction, while IsoDDE advances capabilities further. The Isomorphic Labs raised \$600 million (March 2025), hired a Chief Medical Officer, and is preparing first-in-human oncology trials for early 2026. Combined Lilly and Novartis milestones approach \$3 billion. TxGemma's open-source availability lets pharma companies build custom models on Google's foundation.

OpenAI leads in enterprise LLM deployment through ChatGPT Enterprise, with Moderna, Sanofi, Merck, AstraZeneca, and Thermo Fisher as major pharma customers. Their October 2025 deal with Thermo Fisher embeds OpenAI APIs into clinical research platforms spanning 700+ clinical trials. OpenAI's strength is general-purpose reasoning; their weakness is no pharma-specific model.

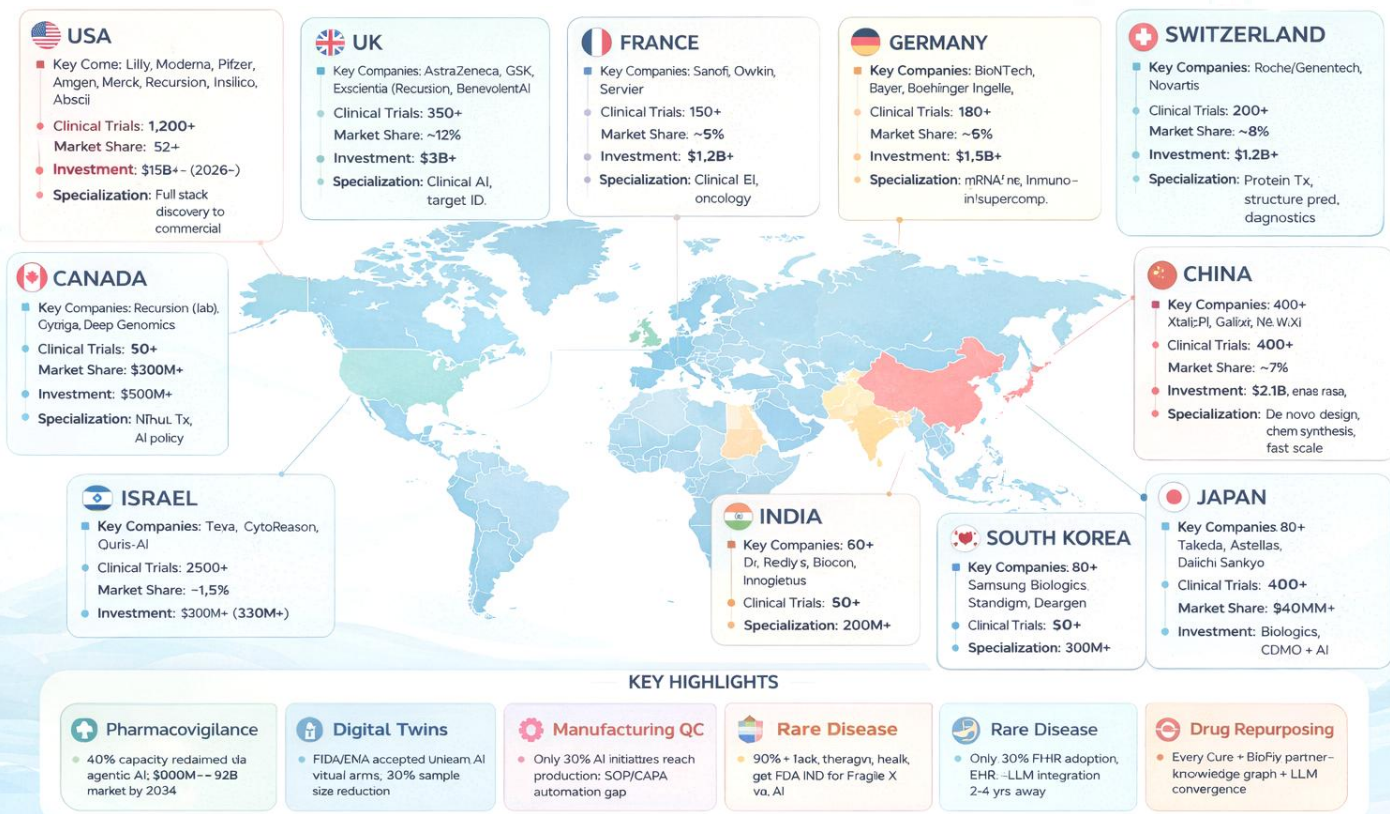
Anthropic is making the most aggressive pharma-specific play, with Claude for Life Sciences, Claude for Healthcare (HIPAA-ready, January 2026), integrations with Benchling and Medidata, and the Coefficient Bio acquisition. Veeva Systems — holding ~80% pharma CRM market share — chose Claude as its AI backbone. Constitutional AI's lower hallucination rates appeal to regulated environments.

Specialized platforms round out the ecosystem. Veeva Systems dominates commercial operations with Vault CRM and regulatory information management. IQVIA provides precision analytics with its "Healthcare-grade AI" and ADA next-best-action engine across 400+ global customers. Certara's CoAuthor cuts regulatory writing first-draft time by 30%. Salesforce Life Sciences Cloud (GA October 2025) has attracted 70+ life sciences companies including Pfizer and Takeda, directly competing with Veeva for pharma CRM.

Pharma Segment	Leading Provider
Drug discovery / structural biology	Google DeepMind/Isomorphic, NVIDIA BioNeMo, EvolutionaryScale
Enterprise AI operations	OpenAI, Anthropic Claude, Microsoft Azure
Clinical trials	AWS, Salesforce, IQVIA, Medidata
Regulatory / medical writing	Certara CoAuthor, Anthropic Claude, Veeva RIM
Commercial / CRM	Veeva (~80% share), Salesforce, IQVIA OCE
GPU infrastructure	NVIDIA (dominant, no peer)
Drug discovery / structural biology	Google DeepMind/Isomorphic, NVIDIA BioNeMo, EvolutionaryScale

Global AI-Driven Drug Discovery: Mapping Innovation, Investment, and Specialization

Which nations lead in AI-driven drug discovery — with hard data



The global landscape of AI-driven drug discovery is rapidly evolving, with significant contributions from leading economies such as the United States, United Kingdom, China, and key European nations. The United States dominates the ecosystem with extensive clinical trials, high investment levels exceeding \$15 billion, and a strong presence of pharmaceutical giants and AI-first biotech companies. Meanwhile, the UK and Europe are carving niches in clinical AI, oncology, and mRNA technologies, supported by steady funding and collaborative research environments.

Asia is emerging as a powerful force in this domain, particularly China, Japan, South Korea, and India. China stands out with over 400 AI-driven companies and strong capabilities in de novo drug design and large-scale chemical synthesis. Japan and South Korea are advancing in biologics, CDMO (Contract Development and Manufacturing Organization), and AI integration, while India is steadily building momentum with a growing number of companies and expertise in generics, biosimilars, and scalable innovation.

Despite the progress, key challenges remain across the global ecosystem. Only a fraction of AI initiatives successfully transition into production, highlighting gaps in manufacturing quality control and regulatory alignment. Opportunities lie in areas such as pharmacovigilance, digital twins, rare disease research, and drug repurposing, where AI can significantly accelerate outcomes. As investment continues to grow and technologies mature, the convergence of AI with biotechnology is set to redefine the future of drug discovery worldwide.

Where the biggest whitespaces remain

Despite rapid adoption, several high-value areas remain underpenetrated by LLMs, representing the most significant opportunities for 2026–2028.

Pharmacovigilance is transitioning fastest. Agentic AI is already reclaiming up to 40% of PV team capacity, and the market is projected to grow from ~\$600 million (2024) to ~\$2 billion by 2034. Tech Mahindra and NVIDIA launched an agentic PV solution in 2025, while France's ANSM integrated AI with AUC ≈ 0.97 for adverse drug reaction detection. Yet NLP-driven PV still harvests only 40% more safety signals than rule-based engines — leaving substantial room for LLM-powered systems to address the 94% median underreporting rate for adverse events.

Real-world evidence generation and personalized medicine remain 2–4 years from mature integration. LLMs combined with knowledge graphs show promise for rare disease detection in EHR data, but data fragmentation, inconsistent coding, and limited regulatory acceptance constrain deployment. Only 30% of U.S. healthcare entities have adopted FHIR interoperability standards.

Manufacturing quality control is a largely untapped LLM whitespace — current AI in manufacturing is dominated by traditional ML and vision systems, with only ~30% of AI manufacturing initiatives reaching full production deployment. LLMs could transform SOP management, deviation investigation, and CAPA documentation, but GxP validation requirements and legacy system integration create 2–4 year timelines.

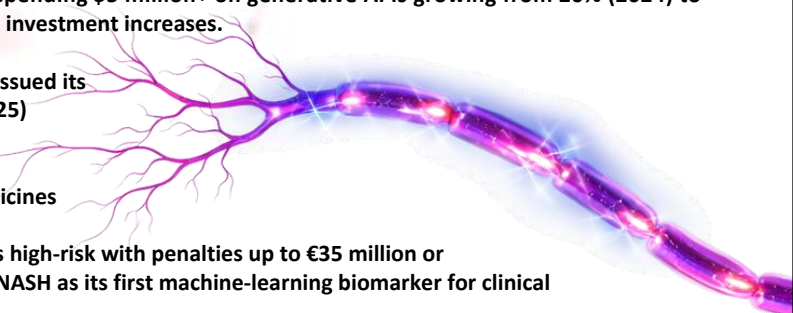
Digital twins for clinical trials represent one of the most active emerging areas. Unlearn.AI's TwinRCTs has been accepted by both FDA and EMA for virtual control arms in Alzheimer's and ALS trials. Sanofi demonstrated blind prediction of Phase 1b asthma trial results using digital twins. The technology may reduce Phase II sample sizes meaningfully while maintaining statistical power and cut preclinical testing times by 30%.

Patient engagement, rare disease research, and drug repurposing remain substantially underexplored. Healx received FDA IND approval for an AI-discovered treatment for Fragile X Syndrome — one of the first AI-driven rare disease breakthroughs. Over 90% of rare diseases lack therapies, and AI's ability to work with sparse datasets through synthetic data augmentation makes this a transformative opportunity.

What 2026–2030 looks like: forecasts and regulatory milestones

Market projections converge on explosive growth. The AI-in-pharma market is projected to reach \$16.5–35 billion by 2031–2034 depending on scope definition. The generative AI segment specifically in drug discovery grows from ~\$319 million (2025) to ~\$2.85 billion by 2034 at 27.4% CAGR. Pharma firms currently dedicate 8–15% of R&D budgets to AI, projected to reach 20–25% by 2030. McKinsey reports that the share of companies spending \$5 million+ on generative AI is growing from 20% (2024) to 32% (2025), with more than two-thirds planning significant investment increases.

The regulatory framework is crystallizing rapidly. The FDA issued its first draft guidance on AI in drug development (January 2025) introducing a 7-step credibility assessment framework, with final guidance expected Q2 2026. The FDA-EMA published 10 joint guiding principles for AI in the medicines lifecycle (January 2026). The EU AI Act took effect in August 2024, classifying many pharma LLM deployments as high-risk with penalties up to €35 million or 7% of global turnover. A landmark: the FDA qualified AIM-NASH as its first machine-learning biomarker for clinical trials (December 2025).



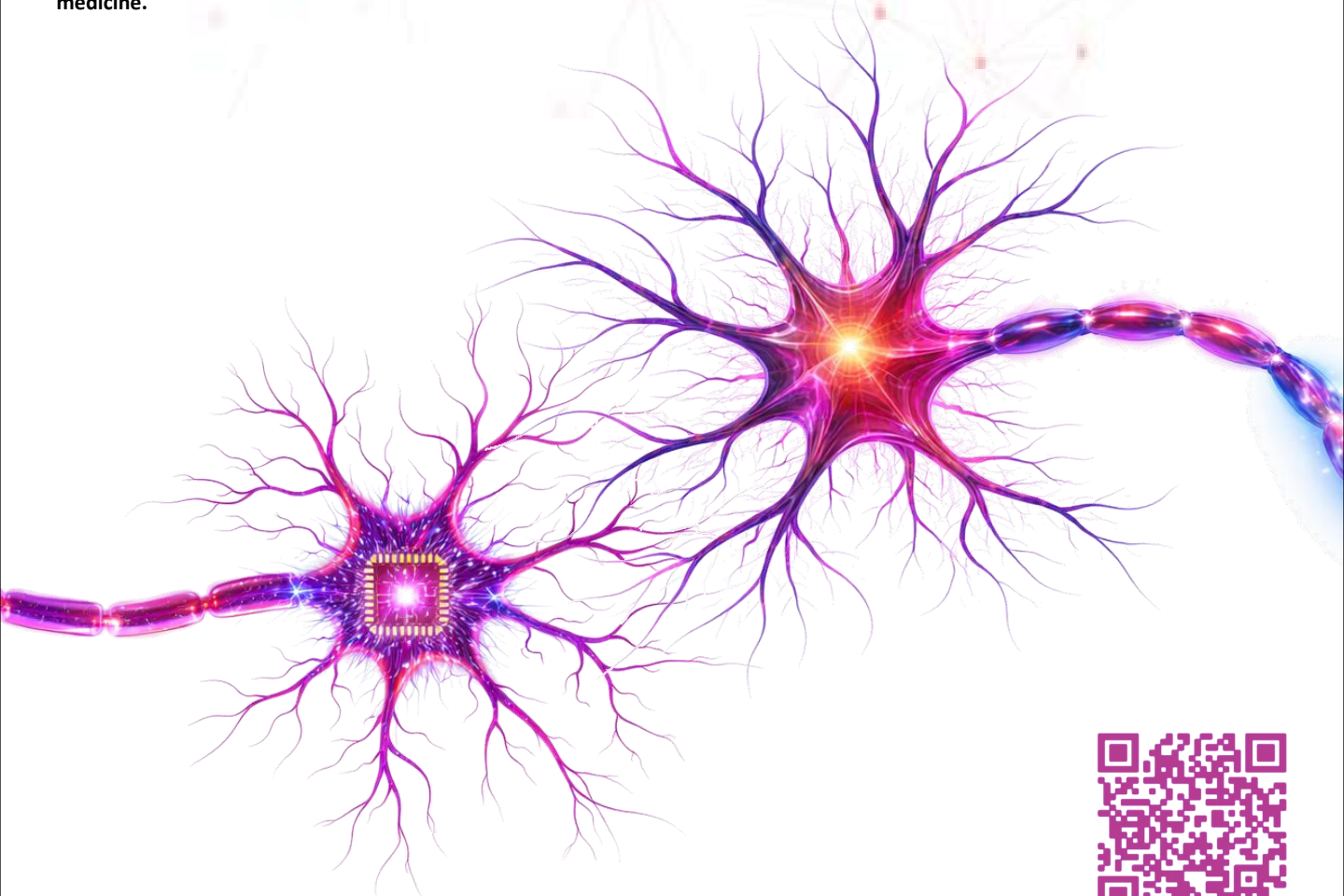
Key predictions for 2026–2030:

- The first AI-discovered drug is expected to receive FDA approval by 2027, with Insilico's rentosertib and Schrödinger-designed NDI-034858 (acquired by Takeda for \$4 billion) as leading candidates
- 2026 is being called "the year of the agent" as agentic AI shifts from pilots to measurable R&D impact across pharma
- Isomorphic Labs will begin first-in-human oncology trials of AI-designed drugs in 2026
- Over 200 AI-assisted drugs are expected to receive regulatory submissions by 2030
- China committed RMB 15 billion (\$2.1 billion) to AI-pharma consortia in 2025, signaling intensifying global competition

Conclusion

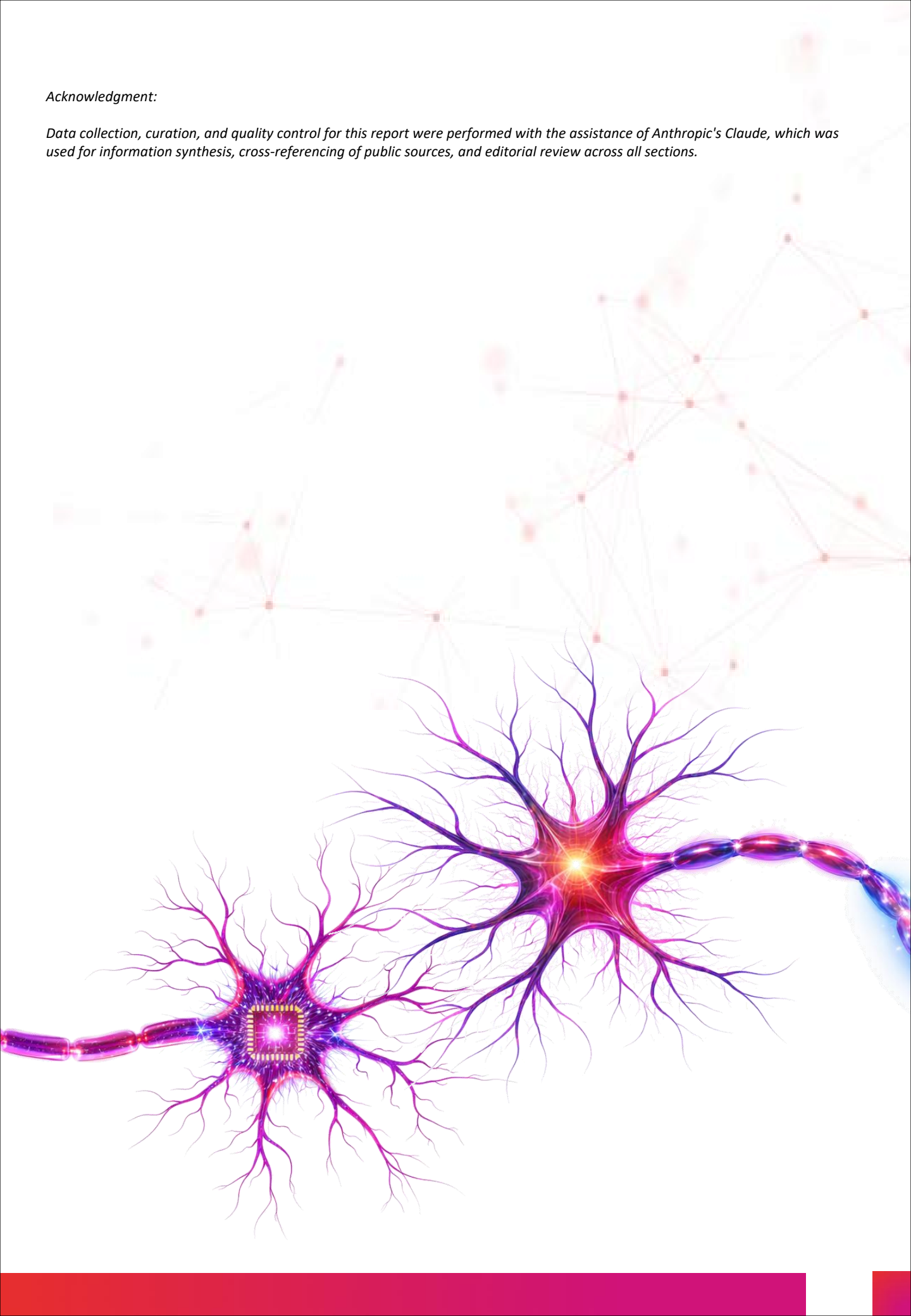
The pharma-LLM landscape in 2025–2026 reveals a field at a decisive inflection point. Investment has reached critical mass — Eli Lilly alone committed over \$5 billion to AI partnerships in 18 months — and enterprise adoption is near-universal among top-20 companies. The technology stack has matured from general-purpose chatbots to 98-billion-parameter protein language models, diffusion-based structure predictors that outperform physics-based methods, and agentic AI systems automating pharmacovigilance and regulatory writing.

Yet the field's ultimate validation — an FDA-approved AI-discovered drug — remains ahead, likely arriving by 2027. The largest whitespace is not in the technology itself, but in the biological complexity and regulatory rigor that govern clinical development. Companies that integrate LLMs deepest into their end-to-end discovery-to-market pipelines — rather than deploying them as isolated productivity tools — will capture disproportionate value. The paradigm is shifting from "search and discover" to "design and create," and the organizations that embrace this shift structurally, not just experimentally, will define the next era of medicine.



Acknowledgment:

Data collection, curation, and quality control for this report were performed with the assistance of Anthropic's Claude, which was used for information synthesis, cross-referencing of public sources, and editorial review across all sections.



References

1. Xu Z, Ren F, Wang P, et al. "A generative AI-discovered TNIK inhibitor for idiopathic pulmonary fibrosis: a randomized phase 2a trial." *Nature Medicine*, June 3, 2025. <https://www.nature.com/articles/s41591-025-03743-2>
2. Dharmasivam M, Azad MG, Richardson V, et al. "Leading artificial intelligence-driven drug discovery platforms: 2025 landscape and global outlook." *Pharmacological Reviews*, Vol 78, No. 1, January 2026. <https://www.sciencedirect.com/science/article/abs/pii/S0031699725075118>
3. Hayes T, et al. "Simulating 500 million years of evolution with a language model (ESM3)." *Science*, January 2025. <https://pubmed.ncbi.nlm.nih.gov/39818825/>
4. Raina Biosciences. "Deep generative models design mRNA sequences with enhanced translational capacity and stability (GEMORNA)." *Science*, August 2025. <https://www.science.org/doi/10.1126/science.adr8470>
5. mRNA-GPT Authors. "mRNA-GPT: A Generative Model for Full-Length mRNA Design and Optimization." *bioRxiv*, 2026. <https://www.biorxiv.org/content/10.64898/2026.03.31.715707v1.full>
6. DrugLM Authors. "DrugLM: A Unified Framework to Enhance Drug-Target Interaction Prediction." *bioRxiv*, July 2025. <https://www.biorxiv.org/content/10.1101/2025.07.09.657250v1.full.pdf>
7. Bai et al. "Machine Learning-Enabled Drug-Induced Toxicity Prediction." *Advanced Science*, 2025. <https://advanced.onlinelibrary.wiley.com/doi/10.1002/adv.202413405>
8. Oxford Academic. "Computational toxicology in drug discovery: applications of AI in ADMET and toxicity prediction." *Briefings in Bioinformatics*, 2025. <https://academic.oup.com/bib/article/26/5/bbaf533/8276062>
9. Wang et al. "Using AI-generated digital twins to boost clinical trial efficiency in Alzheimer's disease." *Alzheimer's & Dementia: Translational Research*, 2025. <https://alz-journals.onlinelibrary.wiley.com/doi/abs/10.1002/trc2.70181>
10. St. Jude Children's Research Hospital. "AI-informed approach to CAR design enhances bi-specific CAR T cells." July 2025. <https://www.stjude.org/media-resources/news-releases/2025-medicine-science-news/ai-informed-approach-to-car-design-enhances-bi-specific-car-t-cells.html>
11. Nature Communications. "AI-guided CAR designs and targeted pathway modulation to enhance multi-antigen CAR T cell durability." 2025. <https://www.nature.com/articles/s41467-025-68272-5>
12. Stanford Medicine. "AI-powered CRISPR could lead to faster gene therapies." September 2025. <https://med.stanford.edu/news/all-news/2025/09/ai-crispr-gene-therapy.html>
13. Reprocell. "OpenCRISPR-1: The First AI-Designed CRISPR Editor for High-Precision Gene & Cell Therapy." 2025. <https://www.reprocell.com/blog/opencrispr-1-the-first-ai-designed-crispr-editor-for-high-precision-gene-cell-therapy>
14. Oxford Academic. "ADNect: a unified framework for predicting the activity of antibody-drug conjugates." *Briefings in Bioinformatics*, 2025. <https://academic.oup.com/bib/article/26/3/bbaf228/8151285>
15. Nature. "Leveraging AI in antibody-drug conjugate development: from target identification to clinical translation in oncology." *npj Precision Oncology*, 2025. <https://www.nature.com/articles/s41698-025-01159-2>
16. PubMed Central. "PepINVENT: generative peptide design beyond natural amino acids." 2025. <https://pubmed.ncbi.nlm.nih.gov/articles/PMC12002334/>
17. Nature. "Optimizing drug design by merging generative AI with a physics-based active learning framework." *Communications Chemistry*, 2025. <https://www.nature.com/articles/s42004-025-01635-7>
18. PubMed Central. "The potential of artificial intelligence in clinical trials." 2025. <https://pubmed.ncbi.nlm.nih.gov/articles/PMC12949343/>
19. PubMed Central. "From Lab to Clinic: How AI is Reshaping Drug Discovery Timelines and Industry Outcomes." 2025. <https://pubmed.ncbi.nlm.nih.gov/articles/PMC12298131/>
20. PubMed Central. "AI in Action: Redefining Drug Discovery and Development." 2025. <https://pubmed.ncbi.nlm.nih.gov/articles/PMC11800368/>
21. MDPI Pharmaceuticals. "Artificial Intelligence in Small-Molecule Drug Discovery: A Critical Review of Methods, Applications, and Real-World Outcomes." 2025. <https://www.mdpi.com/1424-8247/18/9/1271>
22. VUMC (Vanderbilt). "AI can speed antibody design to thwart novel viruses: study (MAGE antibody LM)." November 4, 2025. <https://news.vumc.org/2025/11/04/ai-can-speed-antibody-design-to-thwart-novel-viruses-study/>
23. MedicalXpress. "AI model predicts drug toxicity using differences between animal models and humans." November 2025. <https://medicalxpress.com/news/2025-11-ai-drug-toxicity-differences-animal.html>
24. Nature. "Antibody design enters the AI era." 2024. <https://www.nature.com/articles/d43747-024-00030-w>
25. Pharmaceutical Journal. "Designed by AI: the future of antibody drugs." 2025. <https://pharmaceutical-journal.com/article/feature/designed-by-ai-the-future-of-antibody-drugs>
26. Nature. "Integrating AI into small molecule development for precision cancer immunomodulation therapy." *npj Drug Discovery*, 2025. <https://www.nature.com/articles/s44386-025-00029-y>
27. Fierce Biotech. "Lilly further embraces Insilico's AI tech, inking R&D collab worth up to \$2.75B." March 2026. <https://www.fiercebiotech.com/biotech/lilly-signs-275b-partnership-insilico-ai-engine-pursuit-oral-therapeutics>
28. CNBC. "Eli Lilly reaches \$2.75 billion deal with Insilico to bring AI-developed drugs to global market." March 29, 2026. <https://www.cnbc.com/2026/03/29/eli-lilly-reaches-deal-to-bring-ai-developed-drugs-to-global-market.html>
29. BioPharma Dive. "Lilly's AI commitment expands through deal with Insilico." March 2026. <https://www.biopharmadive.com/news/eli-lilly-insilico-ai-medicine-drug-discovery/816088/>
30. eHealth Magazine. "Insilico Medicine and Servier Forge US\$888 Million Oncology Partnership." January 2026. <https://ehealth.eletsonline.com/2026/01/insilico-medicine-and-servier-forge-us888-million-oncology-partnership-to-advance-ai-led-drug-discovery/>
31. PR Newswire. "Insilico Medicine and Qilu Pharmaceutical Reach Near \$120 Million Drug Development Collaboration." 2026. <https://www.prnewswire.com/news-releases/insilico-medicine-and-qilu-pharmaceutical-ai-reach-near-120-million-drug-development-collaboration-to-accelerate-novel-cardiometabolic-therapies-302671276.html>
32. Insilico Medicine. "Insilico Medicine Announces 2025 Annual Results." 2026. <https://insilico.com/news/ohz9ox0t1-insilico-medicine-announces-2025-annual>
33. pharmparaphorum. "Insilico ends 2025 with \$293m Hong Kong IPO." December 2025. <https://pharmparaphorum.com/news/insilico-ends-2025-293m-hong-kong-ipo>
34. Fierce Biotech. "AI drug designer Insilico Medicine aims to generate nearly \$300M in Hong Kong IPO." December 2025. <https://www.fiercebiotech.com/medtech/ai-drug-designer-insilico-medicine-aims-generate-300m-hong-kong-ipo>
35. BioPharmaTrend. "Insilico Medicine Raises More Than US\$290 Million in Hong Kong's Largest Biotech IPO of 2025." December 2025. <https://www.biopharmatrend.com/news/insilico-medicine-raises-us290-million-in-hong-kongs-largest-biotech-ipo-of-2025-1438/>
36. Fierce Biotech. "Alphabet's Isomorphic stacks two new deals with Lilly, Novartis worth nearly \$3B." January 2025. <https://www.fiercebiotech.com/biotech/alphabets-isomorphic-stacks-two-new-deals-lilly-novartis-worth-nearly-3b-ahead-buzzy-jpm>
37. TechCrunch. "Alphabet's AI drug discovery platform Isomorphic Labs raises \$600M from Thrive." March 31, 2025. <https://techcrunch.com/2025/03/31/alphabets-ai-drug-discovery-platform-isomorphic-labs-raises-600m-from-thrive/>
38. Clinical Trials Arena. "Isomorphic Labs prepares to launch trials for AI-designed drugs." 2026. <https://www.clinicaltrialsarena.com/news/isomorphic-labs-prepares-trials-ai-designed-drugs/>
39. pharmparaphorum. "AI biotech Exscientia and Recursion agree \$688m merger." August 2024. <https://pharmparaphorum.com/news/ai-biotech-exscientia-and-recursion-agree-688m-merger>
40. Fierce Biotech. "Several months after Exscientia merger, AI biotech Recursion reworks pipeline." 2025. <https://www.fiercebiotech.com/biotech/several-months-after-exscientia-merge-ai-outfit-recursion-reworks-pipeline>
41. Recursion Pharmaceuticals (SEC Filing). "Recursion Reports First Quarter 2025 Financial Results." 2025. <https://ir.recursion.com/news-releases/news-release-details/recursion-reports-first-quarter-2025-financial-results-and>
42. Generate:Biomedicines. "Generate:Biomedicines to Initiate Global Phase 3 Studies of GB-0895 for Severe Asthma." December 2025. <https://generatebiomedicines.com/media-center/generatebiomedicines-to-initiate-global-phase-3-studies-of-gb-0895-a-long-acting-anti-tslp-antibody-for-severe-asthma-engineered-with-ai>
43. BioSpace. "Flagship's AI Company Generate Inks Deal Worth up to \$1B with Novartis." 2024. <https://www.biospace.com/deals/flagships-ai-company-generate-inks-deal-worth-up-to-1b-with-novartis>
44. Relay Therapeutics. "Relay Therapeutics Announces Data from Zovagalisib at ESMO TAT Congress 2026." 2026. <https://ir.relaytx.com/news-releases/news-release-details/relay-therapeutics-announces-data-zovagalisib-fulvestrant-phase/>
45. BioSpace. "Relay Therapeutics Reports Full Year 2025 Financial Results and 2026 Milestones." 2026. <https://www.biospace.com/press-releases/relay-therapeutics-reports-fourth-quarter-and-full-year-2025-financial-results-and-outlines-anticipated-2026-milestones>
46. Business Wire. "Chai Discovery Announces \$70 million Series A." August 2025. [https://www.businesswire.com/news/home/20250806670137/en/Chai-Discovery-Announces-\\$70-million-Series-A-To-Transform-Molecular-Design](https://www.businesswire.com/news/home/20250806670137/en/Chai-Discovery-Announces-$70-million-Series-A-To-Transform-Molecular-Design)
47. Absci Corp. "Absci and Owkin Synergize Leading TechBio Platforms." January 2025. <https://investors.absci.com/news-releases/news-release-details/absci-and-owkin-synergize-leading-techbio-platforms-advance/>
48. Absci Corp. "Absci and AMD Announce Collaboration and Strategic Investment." 2025. <https://investors.absci.com/news-releases/news-release-details/absci-and-amd-announce-collaboration-and-strategic-investment>
49. Business Wire. "Unlearn Raises \$50 Million Series C for AI-Powered Digital Twin Technology." February 2024. [https://www.businesswire.com/news/home/20240206806844/en/Unlearn-Raises-\\$50-Million-Series-C-to-Optimize-Clinical-Research-With-AI-Powered-Digital-Twin-Technology](https://www.businesswire.com/news/home/20240206806844/en/Unlearn-Raises-$50-Million-Series-C-to-Optimize-Clinical-Research-With-AI-Powered-Digital-Twin-Technology)
50. Fierce Biotech. "Sanofi, Formation and OpenAI design AI tool (Muse) to slash clinical trial timelines." 2025. <https://www.fiercebiotech.com/cro/sanofi-formation-and-openai-design-ai-tool-slash-clinical-trial-timelines>
51. XtalPI. "XtalPI and DoveTree Announce Landmark \$6 Billion AI Drug Discovery Collaboration." August 2025. <https://en.xtalpi.com/xtalpi-and-dovetree-announce-landmark-6-billion-ai-drug-discovery-collaboration/>
52. PR Newswire. "XtalPI and Pfizer Expand Strategic Collaboration." June 2025. <https://www.prnewswire.com/news-releases/xtalpi-and-pfizer-expand-strategic-collaboration-to-advance-ai-driven-drug-discovery-and-materials-science-simulations-302494033.html>
53. XtalPI. "Signet Therapeutics, Incubated by XtalPI, Nominated for 2025 Prix Gallien USA Award." 2025. <https://en.xtalpi.com/signet-therapeutics-incubated-by-xtalpi-nominated-for-2025-prix-gallien-usa-best-biotechnology-award-for-first-in-class-diffuse-gastric-cancer-drug-sigx1094/>
54. Dyno Therapeutics. "Dyno Launches Three Breakthrough Capsid Delivery Vectors at ASGT 2025." May 2025. <https://www.dynotx.com/news/dyno-therapeutics-launches-three-breakthrough-capsid-delivery-vectors-for-next-generation-eye-medicine-and-cns-gene-therapies-at-the-2025-american-society-of-gene-cell-therapy-asgt-annual-meeting>
55. Business Wire. "Dyno Therapeutics Announces Capsid License Exercised by Astellas." April 2026. <https://www.businesswire.com/news/home/20260408079855/en/Dyno-Therapeutics-Announces-Capsid-License-Exercised-by-Astellas-for-Skeletal-Muscle-Targeted-Gene-Delivery-Validating-AI-Powered-Technology-for-Biological-Sequence-Design>
56. Dyno Therapeutics. "Dyno Announces Exercise of Option by Roche for Next-Generation AAV Vector." January 2025. <https://www.dynotx.com/news/dyno-therapeutics-announces-exercise-of-option-by-roche-for-next-generation-aaav-vector-for-neurological-gene-therapy>
57. PR Newswire. "Somite AI Announces Series A Led by Khosla Ventures." May 2025. <https://www.prnewswire.com/news-releases/somite-ai-announces-series-a-led-by-khosla-ventures-to-revolutionize-cell-therapy-with-ai-foundation-models-302453606.html>
58. Peptilights. "Peptilights Reports Positive 180-Day Top Line Interim Data From Phase 1b Trial of PLG0206." 2025. <https://peptilights.com/news/peptilights-reports-positive-180-day-top-line-interim-data-from-phase-1b-trial-of-plg0206-in-patients-with-periprosthetic-joint-infection-pji/>
59. Debiopharm. "Debiopharm Forges AI-Powered Alliance with NetTargets to Pioneer Dual-Payload ADCs." October 2025. <https://www.debiopharm.com/drug-development/press-releases/debiopharm-forges-ai-powered-alliance-with-nettargets-to-pioneer-dual-payload-adcs-against-drug-resistant-cancers/>
60. Inductive Bio. "Inductive Bio Announces an up to \$21M Award to Develop AI Drug Toxicity Models." December 2025. <https://www.inductive.bio/news/inductive-bio-announces-an-up-to-21m-award-to-develop-ai-drug-toxicity-models>
61. Tempus AI. "Tempus Achieves Record Total Contract Value Exceeding \$1.1 Billion." 2025. <https://www.tempus.com/news/tempus-achieves-record-total-contract-value-exceeding-1-1-billion/>

References

62. Substack (Genomely). "Chan Zuckerberg Biohub Acquires EvolutionaryScale." November 2025. <https://genomely.substack.com/p/chan-zuckerberg-biohub-acquires-evolutionaryscale>
63. SynBioBeta. "EvolutionaryScale Raises \$142M and Unveils AI Model ESM3." June 2024. <https://www.synbiobeta.com/read/evolutionaryscale-raises-142m-and-unveils-ai-model-esm3-to-transform-biology>
64. BenevolentAI. "Baricitinib, First Identified by BenevolentAI As A COVID-19 Treatment, Is Granted Emergency Use In India." 2021. <https://www.benevolent.com/news-and-media/press-releases-and-in-media/baricitinib-first-identified-benevolentai-covid-19-treatment-granted-emergency-use-india-response-its-escalating-crisis/>
65. Privsource. "Osaka Holdings Acquires BenevolentAI and Takes Company Private." March 2025. <https://www.privsource.com/acquisitions/deal/osaka-holdings-acquires-benevolentai-and-takes-company-private-bWSeG2>
66. BioSpace. "Schrodinger Announces Multi-Target Collaboration with Novartis." 2024. <https://www.biospace.com/press-releases/schrodinger-announces-multi-target-collaboration-and-expanded-software-licensing-agreement-with-novartis>
67. Roche. "Roche launches NVIDIA AI factory to accelerate the development of new therapeutics." March 2026. <https://www.roche.com/media/releases/med-cor-2026-03-16>
68. Anthropic. "Advancing Claude in healthcare and the life sciences." 2025. <https://www.anthropic.com/news/healthcare-life-sciences>
69. Anthropic. "Claude for Life Sciences." October 2025. <https://www.anthropic.com/research/claude-for-life-sciences>
70. BioSpace. "AI Giant Anthropic Leans Into Life Sciences With \$400M Coefficient Bio Catch." April 2026. <https://www.biospace.com/business/ai-giant-anthropic-leans-into-life-sciences-with-400m-coefficient-bio-catch>
71. Grand View Research. "Artificial Intelligence in Drug Discovery Market Report, 2033." 2025. <https://www.grandviewresearch.com/industry-analysis/artificial-intelligence-drug-discovery-market>
72. Precedence Research. "AI in the Pharmaceutical Market Size 2025 to 2034." 2025. <https://www.precedenceresearch.com/ai-in-pharmaceutical-market>
73. Mordor Intelligence. "AI in Pharmaceutical Market Analysis & Forecast Report 2031." 2025. <https://www.mordorintelligence.com/industry-reports/artificial-intelligence-in-pharmaceutical-market>
74. Towards Healthcare. "Generative AI in Drug Discovery Market 2025 Soars \$318.55 Bn at 27.42% CAGR." 2025. <https://www.towardshealthcare.com/insights/generative-ai-in-drug-discovery-market-sizing>
75. McKinsey & Company. "Rewiring pharma's regulatory submissions with AI and zero-based design." 2025. <https://www.mckinsey.com/industries/life-sciences/our-insights/rewiring-pharmas-regulatory-submissions-with-ai-and-zero-based-design>
76. McKinsey & Company. "Agentic AI: Unlocking peak performance in biopharma development." 2025. <https://www.mckinsey.com/industries/life-sciences/our-insights/the-synthesis/agentic-ai-unlocking-peak-performance-in-biopharma-development>
77. McKinsey & Company. "AI budgets grow in life sciences." 2025. <https://www.mckinsey.com/featured-insights/week-in-charts/ai-budgets-grow-in-life-sciences>
78. McKinsey & Company. "Generative AI in the pharmaceutical industry: Moving from hype to reality." 2024. <https://www.mckinsey.com/industries/life-sciences/our-insights/generative-ai-in-the-pharmaceutical-industry-moving-from-hype-to-reality>
79. All About AI. "AI in Drug Development Statistics 2026: The \$60 Billion Reality vs. Hype Analysis." January 2026. <https://www.allaboutai.com/resources/ai-statistics/drug-development/>
80. CB Insights. "Pharma AI readiness: How the 50 largest companies by market cap stack up." 2025. <https://www.cbinsights.com/research/ai-readiness-index-pharma-2025/>
81. Drug Target Review. "AI in drug discovery: 2025 in review." 2025. <https://www.drugtargetreview.com/article/192951/ai-in-drug-discovery-2025-in-review/>
82. Ardigen. "AI in Biotech: 2026 Drug Discovery Trends." February 2026. <https://ardigen.com/ai-in-biotech-lessons-from-2025-and-the-trends-shaping-drug-discovery-in-2026/>
83. Coherent Solutions. "Machine Learning and AI in Clinical Trials: Use Cases [2025]." 2025. <https://www.coherentsolutions.com/insights/role-of-ml-and-ai-in-clinical-trials-design-use-cases-benefits>
84. FounderNest. "Pharma Investment Trends 2024-2025: AI to GLP-1." 2025. <https://www.founderest.com/insights/pharma-at-an-inflection-point>
85. Labiotech.eu. "12 AI drug discovery companies you should know about in 2025." 2025. <https://www.labiotech.eu/best-biotech/ai-drug-discovery-companies/>
86. BiopharmaTrend. "How Big Pharma Adopts AI To Boost Drug Discovery." 2025. <https://www.biopharmatrend.com/tech-giants/biopharmas-hunt-for-artificial-intelligence-who-does-what-34/>
87. FDA. "FDA Proposes Framework to Advance Credibility of AI Models Used for Drug and Biological Product Submissions." January 2025. <https://www.fda.gov/news-events/press-announcements/fda-proposes-framework-advance-credibility-ai-models-used-drug-and-biological-product-submissions>
88. European Medicines Agency (EMA). "EMA and FDA set common principles for AI in medicine development." January 2026. <https://www.ema.europa.eu/en/news/ema-fda-set-common-principles-ai-medicine-development-0>
89. FDLI. "Regulating the Use of AI in Drug Development: Legal Challenges and Compliance Strategies." July 2025. <https://www.fdlil.org/2025/07/regulating-the-use-of-ai-in-drug-development-legal-challenges-and-compliance-strategies/>
90. GMP Compliance. "New FDA-Guidance for Industry draft on AI." 2025. <https://www.gmp-compliance.org/gmp-news/new-fda-guidance-for-industry-draft-on-ai>
91. Insilico Medicine — PandaOmics Platform. <https://pharma.ai/pandaomics>
92. Isomorphic Labs — Our Tech. <https://www.isomorphiclabs.com/our-tech>
93. Isomorphic Labs — Partnerships. <https://www.isomorphiclabs.com/partnerships>
94. EvolutionaryScale — ESM3 Release Blog. <https://www.evolutionaryscale.ai/blog/esm3-release>
95. Schrodinger — Pipeline of Drug Discovery Programs. <https://www.schrodinger.com/pipeline/>
96. Generate:Biomedicines — Pipeline. <https://generatebiomedicines.com/pipeline>
97. Absci — Partnerships. <https://www.absci.com/partnerships/>
98. Unlearn.AI — Digital Twins. <https://www.unlearn.ai/digital-twins>
99. Cyagen. "AI-Driven AAV Discovery for CNS & Ophthalmic Gene Therapy." <https://www.cyagen.com/cro-services/ai-powered-aav-discovery>
100. Recursion (SEC Filing). "Recursion Pharmaceuticals 10-K Report." 2025. <https://www.sec.gov/Archives/edgar/data/0001601830/000160183025000034/rxr2025corporatedeck.htm>
101. Schrodinger (SEC Filing). "Schrodinger Q4 2025 Earnings." 2026. <https://www.sec.gov/Archives/edgar/data/0001490978/000149097826000009/sdgr-20251231xxex991.htm>
102. Generate Biomedicines (SEC Filing). "Generate Bio Form 424B4 — IPO Filing." 2026. https://www.sec.gov/Archives/edgar/data/0002100782/000210078226000000/generate_bio_424b4.htm
103. Humai. "AI Drugs Are Already in Clinical Trials — This Is the Most Important Test for Medicine." 2025. <https://www.humai.blog/ai-drugs-are-already-in-clinical-trials-this-is-the-most-important-test-for-medicine/>
104. Humai. "AI-Discovered Drugs Reach Phase III. And 2026 Will Determine Whether All the Promises Were Real." 2026. <https://www.humai.blog/ai-discovered-drugs-reach-phase-iii-and-2026-will-determine-whether-all-the-promises-were-real/>
105. PharmaVoice. "AI deals are surging, and Recursion's ongoing Exscientia merger points to a new world of integration." 2025. <https://www.pharmavoice.com/news/ai-deal-pharma-exscientia-recursion-drug/741040/>
106. C&EN (ACS). "How AI is taking over every step of drug discovery." October 2025. <https://cen.acs.org/physical-chemistry/computational-chemistry/AI-taking-over-step-drug/103/web/2025/10>
107. AVEVA. "When will AI be creating all new drugs." 2025. <https://www.aveva.com/en/our-industrial-life/type/article/when-will-ai-be-creating-all-new-drugs/>
108. Fortune. "Why big pharma is teaming up with AI giants." November 19, 2025. <https://fortune.com/2025/11/19/why-big-pharma-is-teaming-up-with-ai-giants-to-speed-up-drug-discovery-and-make-work-easier-for-health-care-workers/>
109. Singularity Hub. "AI-Designed Antibodies Are Racing Toward Clinical Trials." January 13, 2026. <https://singularityhub.com/2026/01/13/ai-designed-antibodies-race-toward-clinical-trials/>
110. Phys.org. "Robotic microfluidic platform brings AI to lipid nanoparticle design." March 2026. <https://phys.org/news/2026-03-robotic-microfluidic-platform-ai-lipid.html>
111. American Healthcare Leader. "AI Helps Pfizer Write the Future of Bringing Drugs to Market." 2024. <https://americanhealthcareleader.com/2024/ai-brings-drugs-to-market/>
112. World Pharma Today. "Pharma 4.0 and Digital Manufacturing Trends Ahead." 2025. <https://www.worldpharmatoday.com/biopharma/the-rise-of-pharma-4-0-digital-twins-ai-and-predictive-manufacturing/>
113. Yahoo Finance. "Tempus AI: Revenue Mix and 2026 Growth Outlook Explained." 2026. <https://finance.yahoo.com/news/tempus-ai-revenue-mix-2026-144100757.html>
114. Yahoo Finance. "Tempus AI Deepens Merck Collaboration." 2025. <https://finance.yahoo.com/news/tempus-ai-deepens-merck-collaboration-080823627.html>
115. BiopharmaAPAC. "The New AI Gold Rush: Western Pharma's Billion-Dollar Bet on Chinese Biotech." 2025. <https://biopharmaapac.com/analysis/31/7239/the-new-ai-gold-rush-western-pharmas-billion-dollar-bet-on-chinese-biotech.html>
116. IntuitionLabs. "ChatGPT Adoption in the Life Sciences Industry." 2025. <https://intuitionlabs.ai/articles/chatgpt-adoption-life-sciences-industry>
117. IntuitionLabs. "GPTeal: Merck's Generative AI Strategy for Pharma R&D." 2025. <https://intuitionlabs.ai/articles/merck-gpteal-ai-pharma-rd>
118. IntuitionLabs. "AI Applications in Pharmacovigilance and Drug Safety." 2025. <https://intuitionlabs.ai/articles/ai-pharmacovigilance-drug-safety>
119. IntuitionLabs. "Veeva AI Agents: Agentic AI for the Life Sciences Industry." 2025. <https://intuitionlabs.ai/articles/veeva-ai-agents-life-sciences>
120. IntuitionLabs. "Generative AI in mRNA Vaccine Development: COVID-19 Case Study." 2025. <https://intuitionlabs.ai/articles/generative-ai-mrna-vaccine-covid19-case-study>